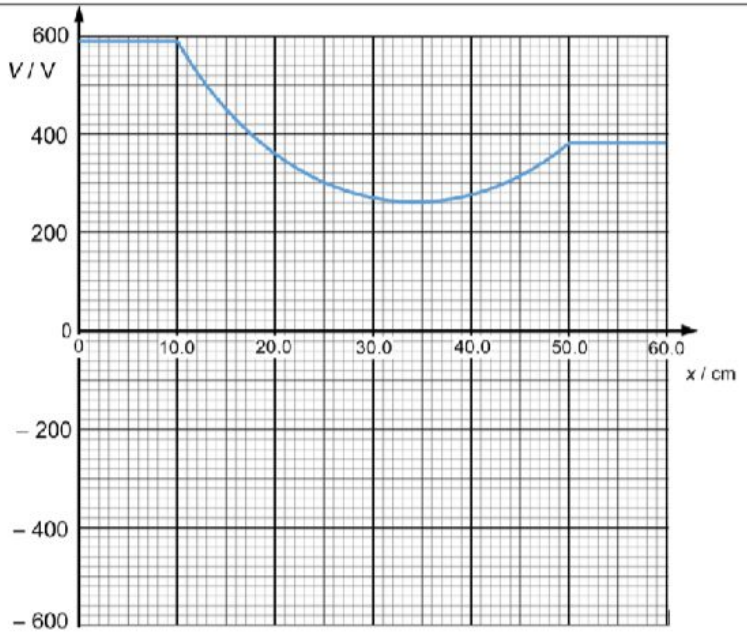
	Contrast between maxima and minima (or fringes) decreases	
6a	field strength (at a point) is equal to potential gradient / gradient of potential-distance graph (at that point) negative sign included or discussed / field strength acts in the direction of decreasing potential	B1 B1
6bi	horizontal solid lines, and non-zero potential within the spheres positive concave curve between A and B minimum nearer to sphere B, i.e. $x > 30.0$ cm values of V_{net} at 10.0 cm and 50.0 cm calculated and plotted with correct unit value of net potential on the surface of A $= \frac{Q_A}{4\pi\varepsilon_o r_A} + \frac{Q_B}{4\pi\varepsilon_o r_B} = \frac{1}{4\pi \times 8.85 \times 10^{-12}}\left(\frac{6.0 \times 10^{-9}}{10 \times 10^{-2}} + \frac{3.0 \times 10^{-9}}{50 \times 10^{-2}}\right) = 593.45 \approx 590$ V value of net potential on the surface of B $= \frac{Q_A}{4\pi\varepsilon_o r_A} + \frac{Q_B}{4\pi\varepsilon_o r_B} = \frac{1}{4\pi \times 8.85 \times 10^{-12}}\left(\frac{6.0 \times 10^{-9}}{50 \times 10^{-2}} + \frac{3.0 \times 10^{-9}}{10 \times 10^{-2}}\right) = 377.66 \approx 380$ V	B1 B1 B1 B1

		
6bii	$E_P = \frac{Qq}{4\pi\varepsilon_o D} = \frac{6.0\times10^{-9}\times3.0\times10^{-9}}{4\pi\times8.85\times10^{-12}\times0.60}$ $E_P = 2.7\times10^{-7}$ J	C1 A1
6biii	Any of the three below – B1) • Both positively-charged/like charges so repel/force is repulsive and move away from each other • EPE of system is converted to KE/force same direction as velocity so speed increases • Total momentum of system is conserved and zero, since the masses are equal so velocities always equal and opposite Force decreases with distance, so acceleration decreases / speed up with decreasing rate	B1 B1
7a	A photon is a quantum of electromagnetic radiation.	B1